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Assignment No. Implementation

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Problem Statement:

The telecommunications sector in Kuwait is experiencing a sharp increase in demand due to technological improvements and people's growing reliance on communication services for both personal and professional purposes. This growing demand is accompanied by a wide range of services and products designed to satisfy clients' ever-changing needs. Telecom firms are constantly diversifying their offerings to maintain their competitiveness in the market, ranging from value-added services and high-speed internet connectivity to basic phone calling. The difficulty of managing the complexity of services, clients, transactions, and staff operations, however, comes along with this increase. Strong database systems that can manage this complexity, guarantee flawless operations, and uphold high customer satisfaction levels are necessary for telecom firms in Kuwait. Such systems are essential for improving service delivery, streamlining workflows, and facilitating well-informed decision-making across organizational levels.

Data Base Telecom Company Description:

We are making a database system for a telecom company based in Kuwait that manages customers phone numbers and services. We've set up a smart and efficient system to keep everything organised and running like a well-oiled machine. It will save customers information and their subscribed services. It offers many services such as purchasing a new number either a normal number or a gold number which is a special number, recharge the number and we have many offers as customers want, they can recharge their phone by 3KD, 5KD, 10KD, 20KD or recharge Internet balance and also, we've many offers start with 5KD, 10KD, 15KD, 20KD.

They can recharge their internet balance monthly or every 6 months. Phone number should be unique. Employees will monitor and manage customers through the telecom services. This system doesn't just keep track of worker details, it is like a big folder holding lots of customer info, from ID numbers to different phone numbers. It is even thinks about families. connecting a kid's number to their parent's information for safety and family-friendly services.

In this company, employees are key players, and their details are really organised in the system. Each employee is identified by a unique employee ID, acting like a special code. The system keeps track of important information like their first name, last name, salary, gender, and phone numbers, making sure everything is in order. The employee ID is like a super important tag, ensuring every employee distinct in the system. These employees play a vital role as they can assist and track services of many customers. As an employee will manage services of many customers and their information should be retained for later use.

While the customers will be offered different services and will have different transactions which will be stored in the database, as the company should keep any invoices done for later problems or legal issues. So, customers information Like age, date of birth, first and last name, civil ID, address, gender, phone numbers will be stored. Also, a customer can have their children's phone number assigned related to them, as customers below 21 cannot have a phone number assigned to themselves. Customer age can be approved by calculating it from the date of birth. Each customer is identified by a unique civil id.

A telecom company have different services for different customers, so each service has different service id, name, description, and price. Each customer can pay and subscribe for more than one service and each service can be managed by employees. Each service is identified by a unique service Id.

Details of every payment will be stored in an invoice in the database like the date of the payment, the invoice id, service name, description, customer phone number, total price, and the name and id and phone number of the employee who managed the payment.so if any problem occurred the details of both parties and transactions will be available. Also, the company make double check every month to make sure there is no mistakes.

This smart system isn't just for the company. It's like a team effort where workers, customers, and services combine smoothly. It's like a well-organised system, making sure everyone enjoys the best telecom experience possible.

Brief Review of the Project and scope:

Our project intends to solve those problems by creating an extensive database system designed especially for a Kuwaiti telecom business. The project's scope includes customer management, service providing, employee administration, and transaction tracking, among other facets of telecom operations.

1- Customer Management:

- The database system will capture and store detailed information about customers, including personal details such as name, age, gender, civil ID, and contact information.
- Special consideration will be given to family-oriented services, where children's phone numbers can be linked to their parents' information for safety and convenience.
- Unique identifiers such as civil IDs will ensure accurate customer identification and enable efficient data retrieval.

2- Service Management:

- The system will catalogue a wide range of telecom services offered by the company, each defined by a unique service ID.
- Service details, including names, descriptions, and pricing structures, will be stored to facilitate service subscription and management.
- Customers will have the flexibility to subscribe to multiple services, and each service can be managed by employees, necessitating efficient service tracking mechanisms.

3- Subscribed Services Management:

- The database system will monitor the services to which clients have paid subscriptions.
- Every subscription will be tracked, including pertinent information like the start and end dates.
- Clients have the option to subscribe to various services, and each subscription will be linked to the specific consumer.
- Different service subscription periods (monthly, annual, etc.) may exist, and the system will manage the renewal process appropriately.
- Billing objectives and ensuring uninterrupted service delivery for clients are the two main uses for subscription details.

4- Employee Management:

- Employee details, including personal information such as names, salaries, and contact details, will be stored in the database.
- Each employee will be assigned a unique employee ID for identification and tracking purposes, ensuring seamless employee management.

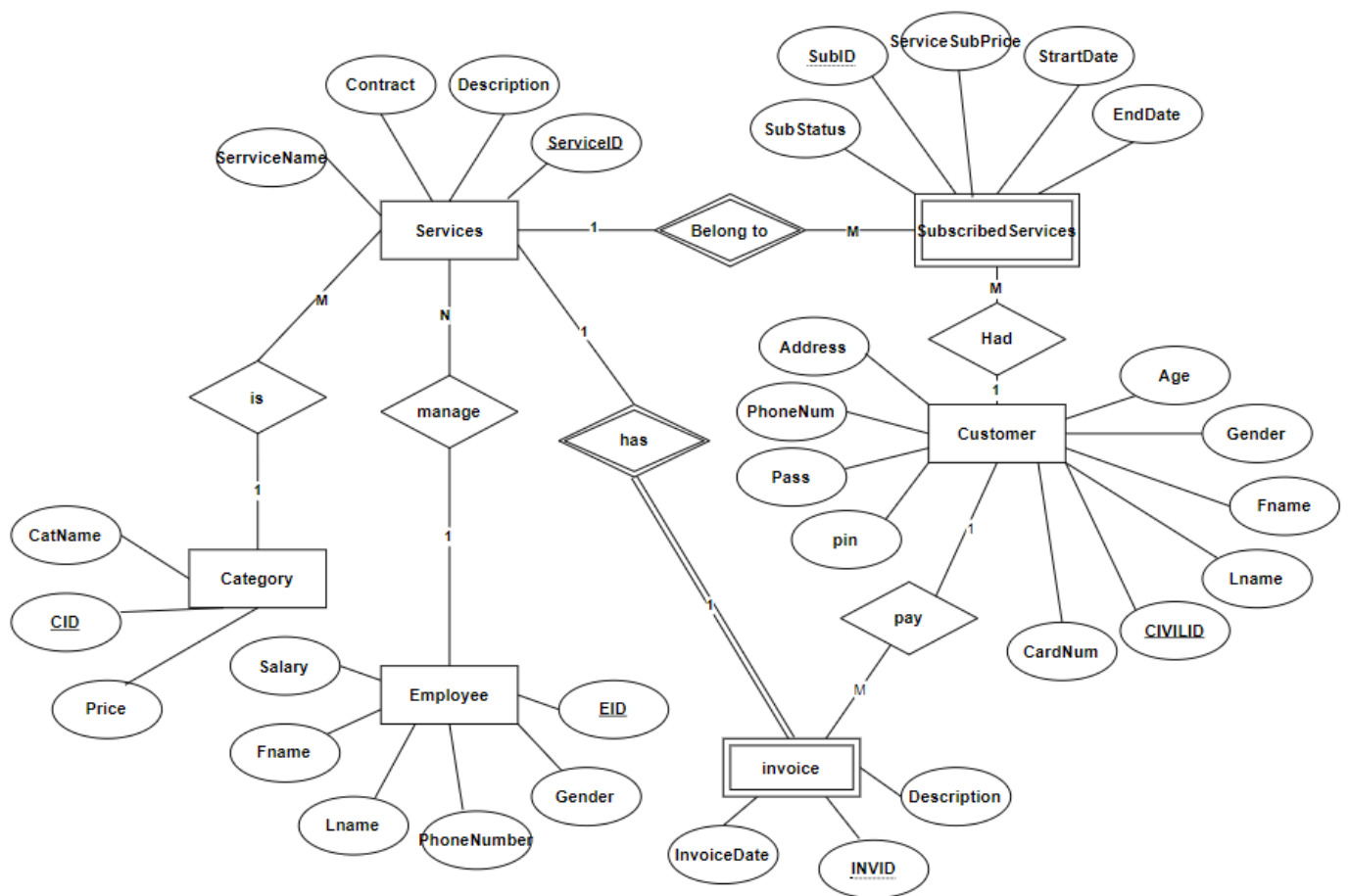
5- Transactions and Invoice Management:

- The system will record and track all customer transactions, including payments, service subscriptions, and invoice generation.
- Detailed invoices will be generated for each transaction, capturing essential information such as transaction dates, invoice IDs, service details, customer and employee information, and total amounts paid.
- Monthly audits will be conducted to verify the accuracy of records and transactions, mitigating the risk of errors, and ensuring regulatory compliance.

Our goal in creating this extensive database system is to give the telecom corporation the resources and skills it needs to effectively resolve the industry's difficulties. In Kuwait's dynamic telecommunications market, the system will work as a centralized platform for managing consumers, services, employees, and transactions, thereby encouraging improved customer satisfaction, operational efficiency, and corporate success.

Database Design:

Entity Relationship Diagram (ER)



Entities and Domains:

Category(CatName, CID, Price)

Domain: CatName as Varchar2(35), CID as number, Price as number.

Services(ServiceName, ServiceID, Contract, Description, CID, EID)

-CID is a foreign key from Category

-EID is a foreign key from employee

Domain: ServiceName as varchar2(35), ServiceID as Number, Contract as Blob, Description as Varchar2(100), CID as number, EID as number.

Employee (EID, Gender, Fname, Lname, PhoneNum, Salary)

Domain: EID as number, gender as varchar2(10), Fname as varchar2(20), Lname as varchar2(20), PhoneNum as varchar2(8) UNIQUE, Salary as number.

Customer (CIVILID, PhoneNum, Address, Fname, Lname, Gender, Age, CardNum, Pass, Pin)

Domain: CIVILID as number, PhoneNumber as varchar2(8) UNIQUE, Address as varchar2(50), Fname as Varchar2(20), Lname as varchar2(20), Gender as varchar2(10), Age as Number, CardNum as number UNIQUE, Pass as number, Pin as number.

SubscribedServices (SubID, StartDate, EndDate, SubStatus, CIVILID, ServiceSubPrice, ServiceID)

- CIVILID is a foreign key from Customer

- ServiceID is a foreign key from Services, while also being a primary key with SubID as SubscribedServices is a weak entity that relies on The Entity Services.

Domain:SubID as number, StartDate as Date, EndDate as Date, SubStatus as varchar2(35), Civilid as number, ServiceSubPrice as Number, ServiceID as number.

Invoice (Description, INVID, InvoiceDate, ServiceID)

- CIVILID is a foreign key from Customer

- ServiceID is a foreign key from Service, while also being a primary key with INVID, as Invoice is a weak entity that depends on Services.

Domain: Description as varchar2(100), INVID as number, InvoiceDate as Date, ServiceID as number.

Relationships:

- one customer can subscribe to many services, a subscribed service can belong to one customer at a time.
- an invoice can belong to one service, and a service have one invoice.
- each subscribed service belongs to a particular service, and each service can have multiple subscribed services.
- an employee can manage multiple services, a service can be managed by one employee at a time.
- a category can have many services, a service can have one category.
- a customer can pay many invoices, an invoice can belong to one customer

Software Design:

1. Customer Management

- Module 1: Add New Customer
- Module 2: Update Customer Information
- Module 3: Delete Customer
- Module 4: View Customer Details

2. Service Management:

- Module 5: Add New Service
- Module 6: Update Service Details
- Module 7: Delete Service
- Module 8: View Service Details

3. Employee Management:

- Module 9: Add New Employee
- Module 10: Update Employee Information
- Module 11: Delete Employee
- Module 12: View Employee Details

4. Subscribed Services Management:

- Module 13: Subscribe to Service
- Module 14: Unsubscribe from Service
- Module 15: View Subscribed Services

5. Transaction and Invoice Management:

- Module 16: Generate Invoice
- Module 17: View Transaction History
- Module 18: Conduct Monthly Audit

Specifications for each module:

Module 1: Add New Customer

- Input: Customer details (name, age, gender, civil ID, address, contact information)
- Output: Success/Failure message
- Algorithm:
 1. Collect input customer details.
 2. Execute SQL query to insert customer details into the Customer table.
 3. Return success message if the insertion is successful; otherwise, return a failure message.

Module 2: Update Customer Information

- Input: Customer ID, updated customer details
- Output: Success/Failure message
- Algorithm:
 1. Collect input customer ID and updated customer details.
 2. Execute SQL query to update the corresponding customer record in the Customer table.
 3. Return success message if the update is successful; otherwise, return a failure message.

Module 3: Delete Customer

- Input: Customer ID
- Output: Success/Failure message
- Algorithm:
 1. Collect input customer ID.
 2. Execute SQL query to delete the corresponding customer record from the Customer table.
 3. Return success message if the deletion is successful; otherwise, return a failure message.

Module 4: View Customer Details

- input: Customer ID
- Output: Customer details
- Algorithm:
 1. Collect input customer ID.
 2. Execute SQL query to retrieve the customer details from the Customer table based on the provided ID.
 3. Display the customer details.

Tables

```
-- Category table
CREATE TABLE Category_p (
    CatName VARCHAR2(35),
    CID NUMBER PRIMARY KEY,
    Price NUMBER
);

-- Employee table
CREATE TABLE Employee (
    EID NUMBER PRIMARY KEY,
    Gender VARCHAR2(10),
    Fname VARCHAR2(20),
    Lname VARCHAR2(20),
    PhoneNum VARCHAR2(8) UNIQUE,
    Salary NUMBER
);

-- Services table
CREATE TABLE Services (
    ServiceName VARCHAR2(35),
    ServiceID NUMBER PRIMARY KEY,
    Contract BLOB,
    Description VARCHAR2(100),
    CID NUMBER,
    EID NUMBER,
    FOREIGN KEY (CID) REFERENCES Category_p(CID),
    FOREIGN KEY (EID) REFERENCES EMPLOYEE(EID)
);

-- Customer table
CREATE TABLE Customer_p (
    CIVILID NUMBER PRIMARY KEY,
    PhoneNum VARCHAR2(8) UNIQUE,
    Address VARCHAR2(50),
    Fname VARCHAR2(20),
    Lname VARCHAR2(20),
    Gender VARCHAR2(10),
    Age NUMBER,
    CardNum NUMBER UNIQUE,
    Pass NUMBER,
    Pin NUMBER
);

-- SubscribedServices table
CREATE TABLE SubscribedServices (
    SubID NUMBER,
    StartDate DATE,
    EndDate DATE,
    SubStatus VARCHAR2(35),
```



```

        CIVILID NUMBER,
        ServiceSubPrice NUMBER,
        ServiceID NUMBER,
        PRIMARY KEY (SubID),
        FOREIGN KEY (CIVILID) REFERENCES Customer_p(CIVILID),
        FOREIGN KEY (ServiceID) REFERENCES Services(ServiceID)
    );

-- Invoice table
CREATE TABLE Invoice (
    Description VARCHAR2(100),
    INVID NUMBER,
    InvoiceDate DATE,
    ServiceID NUMBER,
    CIVILID NUMBER,
    PRIMARY KEY (INVID),
    FOREIGN KEY (ServiceID) REFERENCES Services(ServiceID),
    FOREIGN KEY (CIVILID) REFERENCES Customer_p(CIVILID)
);

```

SQL Queries Needed:

```

--Module4 View customer details --
SELECT * FROM Customer WHERE Civilid = 'Customer ID';

--Module8 View service details --
SELECT * FROM Services WHERE ServiceID = 'input Service ID';

-- Module 12 View employee details --
SELECT * FROM Employee WHERE ID = 'input Employee ID';

-- Module 17 View transaction history --
SELECT * FROM Invoice WHERE ServiceID = 'input Service ID';

```

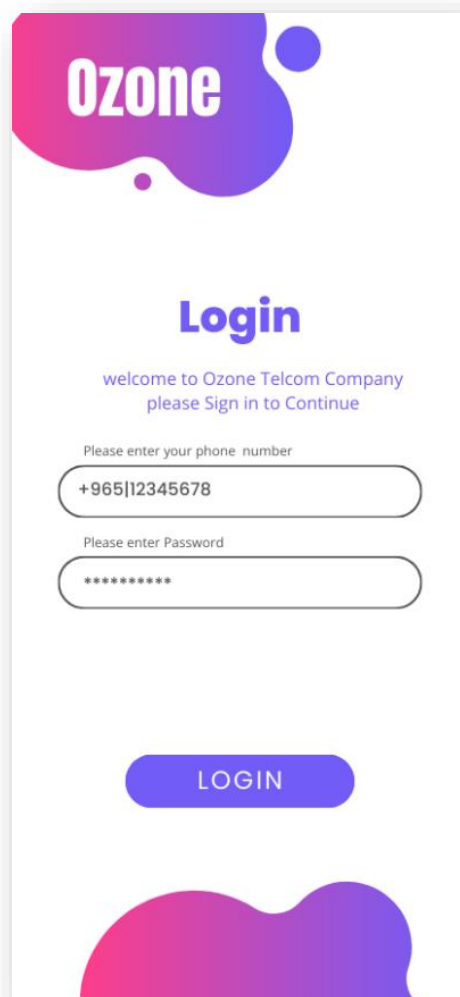
User Interface Design

For this part we will be providing some sketches of the screen that shows how the user interact with the system, while also providing some scenarios that the user goes through to complete a task.

Sketches of the screen

1. Welcome screen.

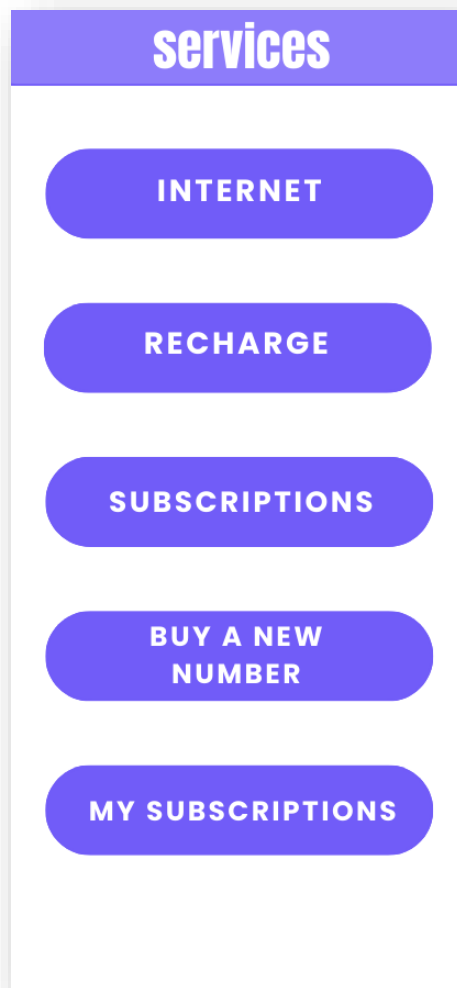
This screen will ask the user to log into their Ozone Telcom account by entering their phone number and password. If log in information is approved, they will be redirected to the main menu.



The image is a vertical rectangular mockup of a mobile application login screen. At the top left, there is a logo consisting of the word "Ozone" in white, bold, sans-serif font, set against a purple and pink gradient bubble shape. Below the logo, the word "Login" is centered in a bold, purple, sans-serif font. Underneath "Login", there is a smaller, purple, sans-serif text that reads "welcome to Ozone Telcom Company" followed by "please Sign in to Continue" on the next line. Below this text, there are two input fields. The first field is labeled "Please enter your phone number" and contains the text "+965|12345678". The second field is labeled "Please enter Password" and contains a series of asterisks "*****". Below the input fields, there is a purple rounded rectangular button with the word "LOGIN" in white, uppercase, sans-serif font. At the bottom of the screen, there is a decorative purple and pink gradient shape.

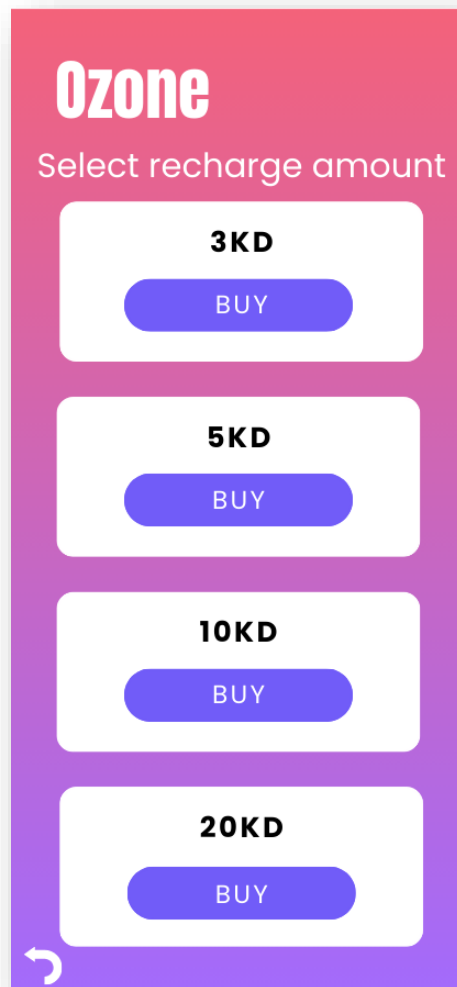
2. Services

Next page which is the main menu will offer the user the various services available for the user to choose from.



3. Service Details

This screen will show more specific details for specified chosen service. e.g. different balance recharge amounts for the user to choose from.



4.Payment Details

To pay for the service the user should fill their payment information and if the payment is successful a message will appear confirming it. Also, other details of the invoice will be shown.

Ozone

Payment Page

Payment for: Recharge Balance

Price: 5 KD

Phone Number: +965| 12345678

Payment Date: 07-03-2024

Invoice to Ozone Telcom company

Please fill in your Billing details

First Name

Last Name

 credit card number

 PIN

CONFIRM



Ozone

Payment Page

Payment for: Recharge Balance

Price: 5 KD

Phone Number: +965| 12345678

Payment Date: 07-03-2024

Invoice to Ozone Telcom company

Balance have been
successfully Recharged

OK

ALMUKHTAR

 xxxxxxxxxxxx

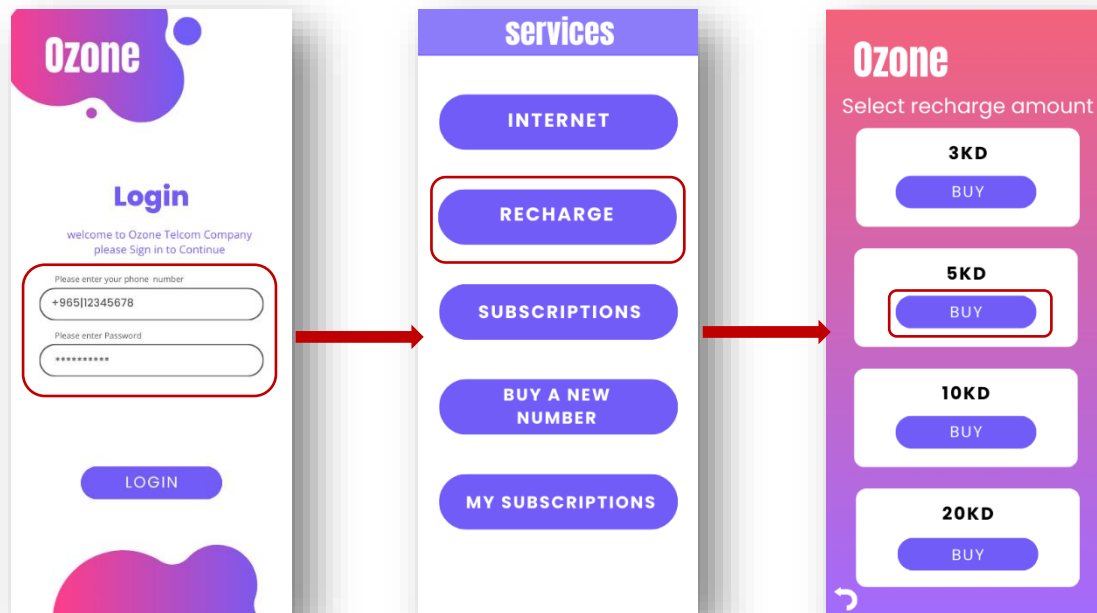
 xxxx

CONFIRM



Scenarios

We will demonstrate a scenario that will explain the different steps the user goes through to do a certain task.



Suppose the user wants to recharge their phone balance with 5 Kuwaiti Dinars (KD)

First, they will open the application which will welcome the user and ask them to enter their information to log into their account to continue.

Next the user will be presented with menu with different services that they can they make use of; in this scenario the user will choose the recharge option because they want to recharge their balance, otherwise the user can go back to the main menu.

This will lead them to another screen which will further specify how much balance the user want to buy and add it to their balance, so the user will choose 5KD like the case of our scenario.

Next user will be directed to a payment page to pay for the required service, otherwise user can go back. If the user filled the required information, then clicked confirm, and the payment was successful a message will pop up showing that it is successful. And the user will have their balance recharged successfully by 5KD as in the scenario.

Ozone Payment Page

Payment for: Recharge Balance
Price: 5 KD
Phone Number: +965| 12345678
Payment Date: 07-03-2024
Invoice to Ozone Telcom company

Please fill in your Billing details

First Name
Last Name
credit card number
PIN

CONFIRM

Ozone Payment Page

Payment for: Recharge Balance
Price: 5 KD
Phone Number: +965| 12345678
Payment Date: 07-03-2024
Invoice to Ozone Telcom company

Please fill in your Billing details

HUDA
ALMUKHTAR
XXXXXXXXXXXX
XXXX

CONFIRM

Ozone Payment Page

Payment for: Recharge Balance
Price: 5 KD
Phone Number: +965| 12345678
Payment Date: 07-03-2024
Invoice to Ozone Telcom company

Balance have been successfully Recharged

OK

CONFIRM

Implementation Plan

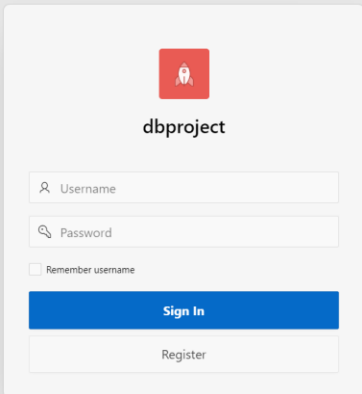
For the implementation of our design, we will be using Oracle SQL developer to maintain and access data. For starters we will be using draw.io to construct our entity Relationship diagram (ER) which will illustrate how the entities of our design relate to each other. Also using Canva to portray the software of the user interface design (UI) in a sketch. For the operating system we will be using apex oracle, while the languages are SQL and HTML.

Apex system will be used to make the user interface as the hardware chosen is computers and laptops. Implementation plan will take about 2 months to complete both database and the operating system . Amal will be working on implementing the database, while I (Huda) will be implementing the user interface. Each of us will be testing for errors at the same time of the implementation. Later on, after both of the database system and the user interface are finished, we will be checking for any possible errors within the implementation of the database and the operating system.

User's Manual

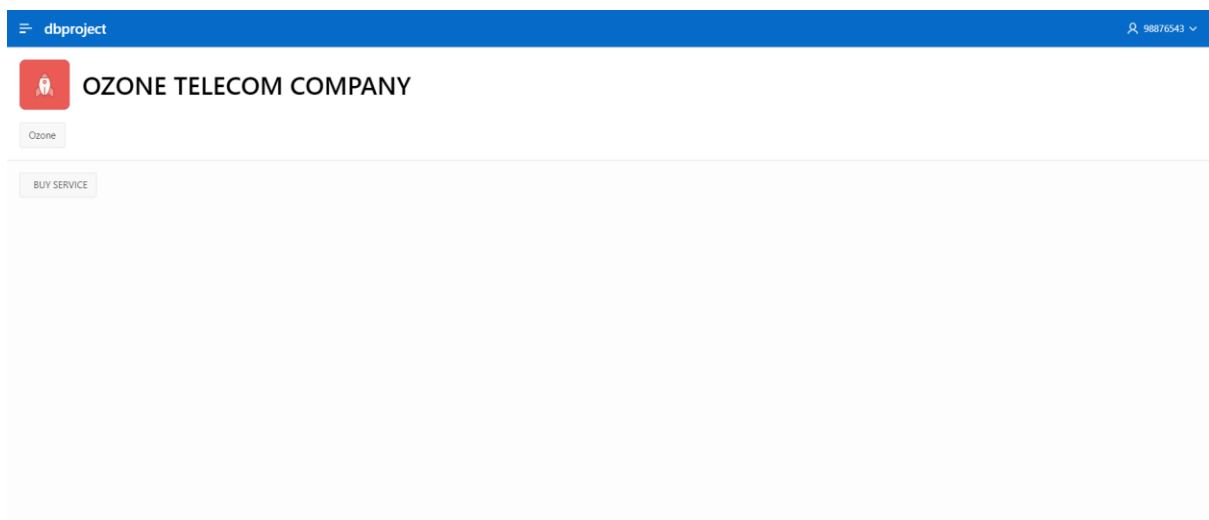
There are two types of users for this system customer and employee. Customers can log in to buy a certain service. Employees can also login to monitor and manage services by checking the customers list, services list, and invoices. Employees can add new services and customers to the system and generate invoices. Also, they can add new employees to the system.

Log in

A login form for 'dbproject' is centered on a light gray background. The form is white with a red square logo containing a white person icon at the top. Below the logo is the text 'dbproject'. The form contains two input fields: 'Username' with a magnifying glass icon and 'Password' with a magnifying glass icon. Below these is a checkbox labeled 'Remember username'. At the bottom of the form are two buttons: a blue 'Sign In' button and a white 'Register' button with a gray border.

First the user will be obligated to sign in by inserting their username (phone number) and password to continue. They will be directed to home page in the next figure.

Home page

The home page of the 'OZONE TELECOM COMPANY' is shown. It features a blue header bar with the 'dbproject' logo on the left and a user profile icon with the number '98876543' on the right. Below the header, there is a red square logo with a white person icon, followed by the text 'OZONE TELECOM COMPANY'. Underneath this, there is a gray button labeled 'Ozone'. At the bottom of the page, there is a gray button labeled 'BUY SERVICE'.

User can choose from available services by clicking on buy services.

This will direct them to the service form.

The screenshot shows the 'Service form' page in the 'dbproject' application. The page has a blue header with the 'dbproject' logo and a user profile icon. Below the header, there is a breadcrumb 'Home \\' and the title 'Service form'. The form itself is a light gray box with the title 'Service form' at the top. It contains several input fields: 'Eid', 'SERVICEID', 'Servicename', 'Contract' (with a 'Choose File' button), 'Description', and 'Cid'. At the bottom of the form, there are 'Cancel' and 'Create' buttons.

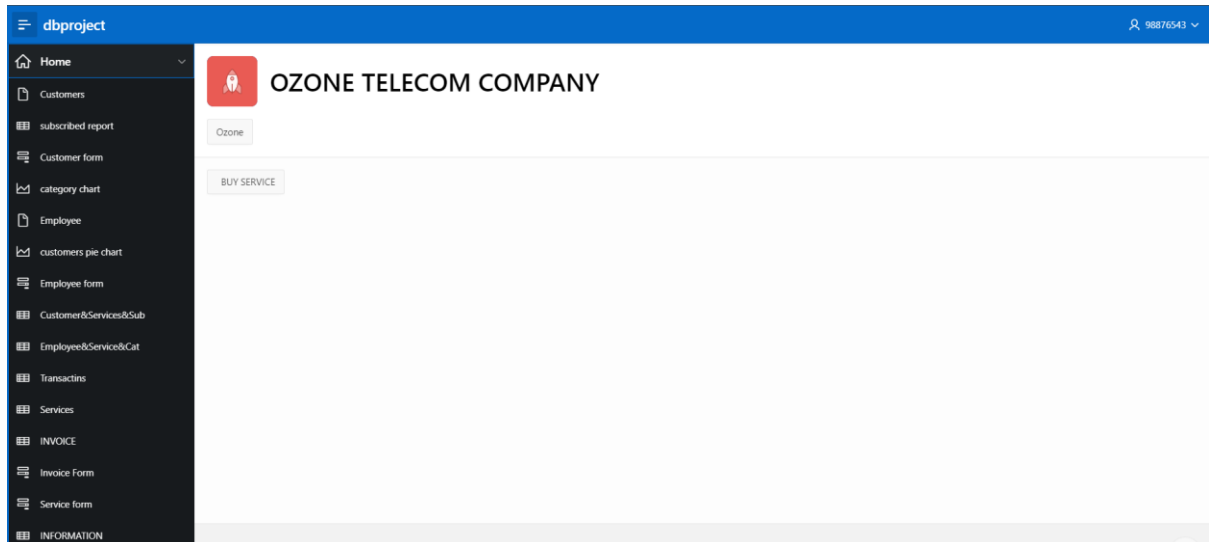
So now the user can fill the form accordingly to add it in the system. Which will direct the user to the invoice form.

The screenshot shows the 'Invoice Form' page in the 'dbproject' application. The page has a blue header with the 'dbproject' logo and a user profile icon. A green notification banner at the top right says 'Row created.' with a close button. On the left, there is a dark sidebar with a menu. The main content area has a breadcrumb 'Home \\' and the title 'Invoice Form'. The form is a light gray box with the title 'Invoice Form' at the top. It contains several input fields: 'INVID', 'Description', 'Invoicedate' (with a calendar icon), 'Serviceid', 'Civild' (a dropdown menu), and 'category name and price' (a dropdown menu). At the bottom of the form, there are 'Cancel' and 'Create' buttons.

After filing the invoice with the customer information (customer must be registered in the system previously) the payment information will be saved in the database. User will be directed to home page again.

menu

Users can also navigate to other pages with other functionality. As user can directly make invoice, new service, new customer, etc. separately from the buy service.



Add customer

You can insert new customers to the database by filling the following form.

The screenshot displays the 'Customer form' page within the 'dbproject' application. The sidebar and header are identical to the previous screenshot. The main content area is titled 'Customer form' and contains a form with the following input fields: 'civild', 'Phonenum', 'Address', 'Fname', 'Lname', 'Gender', 'Age', and 'Cardnum'. Each field is represented by a light gray rectangular input box.

Add employee

New employees can be registered by user.

dbproject 98876543

Home \ Employee form

Employee form

EID

Gender

Fname

Lname

Phonenum

Salary

Pass

Cancel Create

More functionalities

Can check all employees and customers and their information.

dbproject 98876543

Home \ Employee

Employee

Names

Eid ↑	Gender	Fname	Lname	Phonenum	Salary	Pass
1	MALE	MOHAMMAD	AHMAD	96675454	1000	123
2	FEMALE	SARA	ALI	98876543	1000	123
3	MALE	SALEM	ABDULLAH	99976788	1200	123
4	FEMALE	MARIAM	KHALID	94487655	1200	123
6	male	azoz	mohammed	51580601	1000	123

1 - 5

Release 1.0

dbproject

98876543

Home

Customers

subscribed report

Customer form

category chart

Employee

customers pie chart

Employee form

Customer&Services&Sub

Employee&Services&Cat

Transactins

Services

INVOICE

Invoice Form

Service form

INFORMATION

Home \

Customers

Customers

Names

Civilid ↑	Phonenum	Address	Fname	Lname	Gender	Age	Cardnum	Pass	Pin
1122334455	90908765	JABRYA,BLOCK5	LATIFA	TAREQ	FEMALE	30	7878787	12345678	1234
5555555555	56565656	Salwa66	Amal	Aldhafeeri	female	22	9879879	123	3333
5656565656	97680753	JAHRA,BLOCK3	AHMAD	SAUD	MALE	40	6767546	12345678	1234
5656784901	98876765	SALMYA,BLOCK7	NADA	WALEED	FEMALE	27	5667788	12345678	1234
6666666666	66064576	salwa3	mohammed	ayied	male	30	8888888	123	7777
8765483092	99988845	SALWA,BLOCK10	NASER	BADER	MALE	21	8887777	12345678	1234

1 - 6

Can see subscribed services by a specific customer.

dbproject

98876543

Home

Customers

subscribed report

Customer form

category chart

Employee

customers pie chart

Employee form

Customer&Services&Sub

Employee&Services&Cat

Transactins

Services

INVOICE

Invoice Form

Service form

INFORMATION

View subscribed service by a specific customer

subscribed report

Subid ↑	Startdate	Enddate	Substatus	Civilid	Servicesubprice	Serviceid
1	3/24/2024	9/24/2024	ACTIVE	5656784901	60	1

Release 1.0

Chart that shows available services and their prices

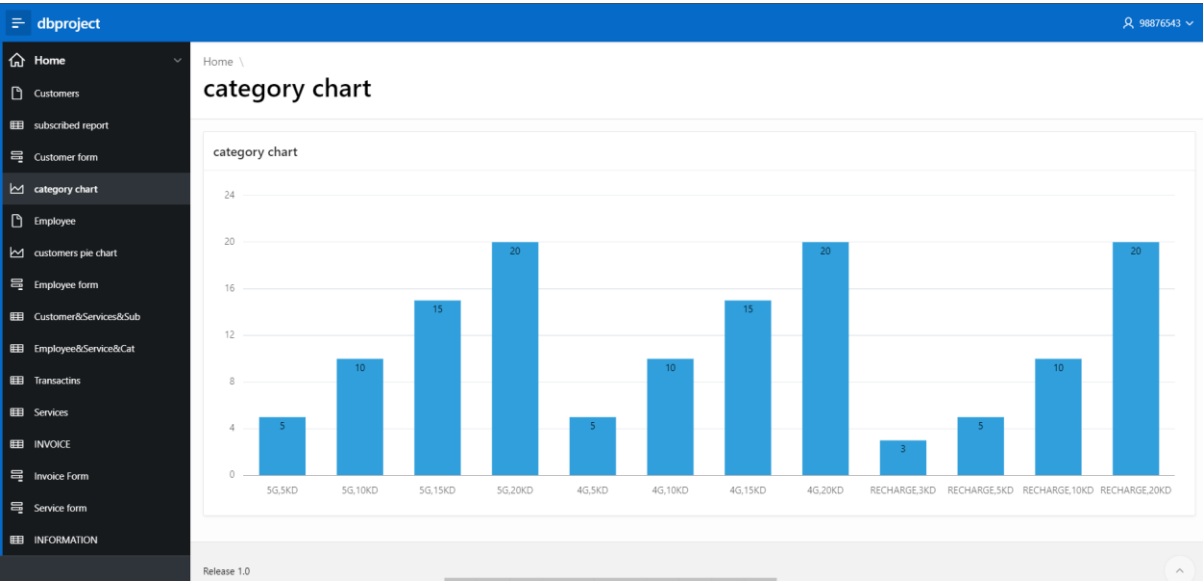
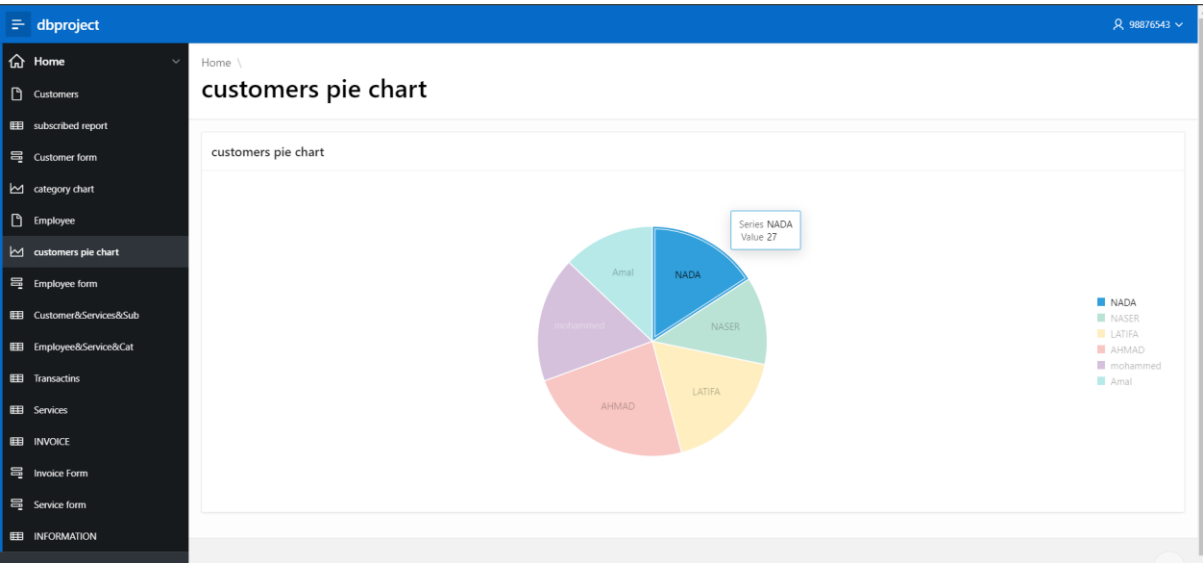


Chart showing different age categories of the customers



Tables with customers services subscriptions

dbproject 98876543

Select the subscribed service IDs, start dates, and end dates along with the first name and last name of the customers and the service names they are subscribed to:

Home \ Customer&Services&Sub

Subid	Startdate	Enddate	Fname	Lname	Servicename
1	3/24/2024	9/24/2024	NADA	WALEED	INTERNET
2	3/25/2024	9/25/2024	NASER	BADER	INTERNET
3	3/26/2024	9/26/2024	LATIFA	TAREQ	INTERNET

Release 1.0

Table for services managed by employees

dbproject 98876543

Select the first name, last name, and salary of employees along with the service names they are responsible for and the corresponding prices:

Home \ Employee&Service&Cat

Fname	Lname	Salary	Servicename	Price
MOHAMMAD	AHMAD	1000	INTERNET	5
MOHAMMAD	AHMAD	1000	INTERNET	5
MOHAMMAD	AHMAD	1000	INTERNET	10
SARA	ALI	1000	RECHARGE	3
SARA	ALI	1000	RECHARGE	3
SARA	ALI	1000	RECHARGE	3
SARA	ALI	1000	RECHARGE	3
SARA	ALI	1000	INTERNET	20
SARA	ALI	1000	INTERNET	20

Release 1.0

Offered Services and their contract and related information.

dbproject

98876543

Home

Customers

subscribed report

Customer form

category chart

Employee

customers pie chart

Employee form

Customer&Services&Sub

Employee&Services&Cat

Transactins

Services

INVOICE

Invoice Form

Service form

INFORMATION

Home \

Services

Serviceid ↑%	Servicename	Contract	Description	Cid	Eid
1	INTERNET		BUY INTERNET 5G BALANCE OF 5KD	1	1
2	INTERNET		BUY INTERNET 4G BALANCE OF 10KD	6	1
3	RECHARGE		RECHARGE NUMBER OF 10KD	9	2
4	INTERNET	Download	BUY INTERNET	1	1
5	INTERNET	Download	BUY INTERNET	8	2
6	INTERNET	Download	INTERNET	8	2
7	RECHARGE	Download	RECHARGE NUMBER	9	2
8	RECHARGE		RECHARGE SKD	9	2
21	RECHARGE	Download	RECHARGE	9	2

Release 1.0

Transactions and invoices

dbproject

98876543

Home

Customers

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Description ↑%	Invid	Invoicedate	Serviceid	Civilid
INTERNET	21	5/6/2024	5	1122334455
INTERNET	22	5/1/2024	4	6666666666
	1	3/24/2024	1	5656784901
	2	3/24/2024	2	5656784901

Release 1.0

Information of specific services descriptions and category and the customer who bought them and the price.

dbproject

98876543

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ServiceName ↑↓	ServiceId	Fname	Lname	Civilid	Description	Price	Cid
INTERNET	1	NADA	WALEED	5656784901	BUY INTERNET 5G BALANCE OF 5KD	5	1
INTERNET	5	LATIFA	TAREQ	1122334455	BUY INTERNET	20	8
INTERNET	2	NADA	WALEED	5656784901	BUY INTERNET 4G BALANCE OF 10KD	10	6
INTERNET	4	mohammed	ayied	6666666666	BUY INTERNET	5	1

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Implementation Manual

The system contains 18 pages in total including the log in page.

First page is the log in page-9999 which will send you to Home page-1 and after choosing the buy service it will redirect you to the services form page-24, after creating the service you will be redirected to the invoice form page-17 after that you will be redirected again to the home page-1.

However, you can access different pages manually by selecting the page from the menu. Those pages act individually and do not necessarily need to direct you to other pages.

Contract is downloadable as a Blob file.

Apex workspace:

WORKSPACE: DBLABKUAMAL

Username: S2202144825@KU.EDU.KW

password: amal1234

project name: dbproject

[dbproject - Log In \(oracle.com\)](#)

<https://apex.oracle.com/pls/apex/r/dblabkuamal/dbproject/login?session=103355976548609>

Code listing

[database SQL code and tables are listed previously]

listing of codes used in oracle apex application pages:

Page 3: `SELECT * FROM SUBSCRIBEDSERVICES WHERE CIVILID = 5656784901`

[Which will View subscribed service by a specific customer]

Page 4:

```
SELECT
SERVICENAME,SERVICES.SERVICEID,CUSTOMER_P.FNAME,CUSTOMER_P.LNAME,CUS
TOMER_P.CIVILID,SERVICES.DESRIPTION,CATEGORY_P.PRICE,CATEGORY_P.CID
FROM CUSTOMER_P,SERVICES,INVOICE,CATEGORY_P
WHERE CUSTOMER_P.CIVILID=INVOICE.CIVILID
AND INVOICE.SERVICEID=SERVICES.SERVICEID
AND CATEGORY_P.CID=SERVICES.CID
```

[Query that shows Information of specific services descriptions and category and the customer who bought them and the price]

Page 10:

```
SELECT SS.SubID, SS.StartDate, SS.EndDate, C.Fname, C.Lname, S.ServiceName
FROM SubscribedServices SS
JOIN Customer_p C ON SS.CIVILID = C.CIVILID
JOIN Services S ON SS.ServiceID = S.ServiceID
```

[Query that Shows subscribed service IDs, start dates, and end dates along with the first name and last name of the customers and the service names they are subscribed to.]

Page 11:

```
SELECT E.Fname, E.Lname, E.Salary, S.ServiceName, C.Price
```

```
FROM Employee E
```

```
JOIN Services S ON E.EID = S.EID
```

```
JOIN Category_p C ON S.CID = C.CID
```

[Query which results in first name, last name, and salary of employees along with the service names they are responsible for and the corresponding prices.]

Page 14:

```
SELECT * FROM Invoice
```

Page 17:

```
select
```

```
"SERVICENAME","SERVICEID",sys.dbms_lob.getlength("CONTRACT")"CONTRACT","DESCRIPTION","CID","EID"from "SERVICES"
```

[Offered Services and their contract and related information.]

Sample Output

Add customer

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Customer form

civild
1234567891

Phonenum
87654321

Address
Andalous, block13

Fname
Abrar

Lname
Ibrahim

Gender
female

Age
29

Cardnum
1122334

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Row created.

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Names

Civild	Phonenum	Address	Fname	Lname	Gender	Age	Cardnum	Pass	Pin
1122334455	90908765	JABRYA,BLOCK5	LATIFA	TAREQ	FEMALE	30	7878787	12345678	1234
1234567891	87654321	Andalous, block13	Abrar	Ibrahim	female	29	1122334	123	123
5555555555	56565656	Salwa66	Amal	Aldhafeeri	female	22	9879879	123	3333
5656565656	97680753	IAHRA,BLOCK3	AHMAD	SAUD	MALE	40	6767546	12345678	1234
5656784901	98876765	SALMYA,BLOCK7	NADA	WALEED	FEMALE	27	5667788	12345678	1234
6666666666	66064576	salwa3	mohammed	ayied	male	30	8888888	123	7777
8765483092	99988845	SALWA,BLOCK10	NASER	BADER	MALE	21	8887777	12345678	1234

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Add employee

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Employee form

EID5

Genderfemale

FnameNoor

LnameOsama

Phonenum90091234

Salary1000

Pass123

Cancel

Create

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Row created.

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Employee

Names

Eid	Gender	Fname	Lname	Phonenum	Salary	Pass
1	MALE	MOHAMMAD	AHMAD	96675454	1000	123
2	FEMALE	SARA	ALI	98876543	1000	123
3	MALE	SALEM	ABDULLAH	99976788	1200	123
4	FEMALE	MARIAM	KHALID	94487655	1200	123
5	female	Noor	Osama	90091234	1000	123
6	male	azoz	mohammed	51580601	1000	123

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Generate invoice

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INTERNET	5/1/2024	4	66064576
	3/24/2024	1	98876765
	3/24/2024	2	98876765

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